CHARACTERISATION OF THE CLASSES C_J OF A FINITE LOCALLY FREE SHEAF BY A SINGLE RELATION ON ITS PROJECTIVE BUNDLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The projective space bundle formula. Same chapter chow.tex, label lemma-chow-ring-projective-bundle, invoked at proof line 7303. Gives the uniqueness of alpha_0, ..., alpha_r with alpha_0 = alpha satisfying the displayed relation, by identifying the Chow group of the projective bundle with the direct sum of the Chow groups of the base.
- Flat pullback versus proper pushforward, and the projection formula. Same chapter chow.tex, labels lemma-flat-pullback-proper-pushforward (proof line 7348) and lemma-pushforward-cap-c1 (proof line 7349). Allows the relation proved on the disjoint union of the projective bundles over the components to be transported back to P, and then to the relation with pi^*(p_* beta_j) on X.

Source reference. Original lemma 02TX: sources/stacks-chow/pinned-original-source.tex, source lines 6932–6932. Navigation pointer for \ref{lemma-chow-ring-projective-bundle} used inside the counted span (line 7303); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RG: sources/stacks-chow/pinned-original-source.tex, source lines 2401–2401. Navigation pointer for \ref{lemma-flat-pullback-proper-pushforward} used inside the counted span (line 7348); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SU: sources/stacks-chow/pinned-original-source.tex, source lines 4631–4631. Navigation pointer for \ref{lemma-pushforward-cap-c1} used inside the counted span (line 7349); the argument that consumes it is retained in the span.

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Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 7281); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{E}$ be a finite locally free sheaf of rank r on X. Let $(\pi : P \to X, \mathcal{O}_P(1))$ be the projective bundle associated to $\mathcal{E}$. For $\alpha \in Z_k(X)$ the elements $c_j(\mathcal{E}) \cap \alpha$ are the unique elements α_j of $\mathrm{CH}_{k-j}(X)$ such that $\alpha_0 = \alpha$ and*

$$\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_P(1))^i \cap \pi^*(\alpha_{r-i}) = 0$$

holds in the Chow group of P.

Proof. The uniqueness of $\alpha_0, \ldots, \alpha_r$ such that $\alpha_0 = \alpha$ and such that the displayed equation holds follows from the projective space bundle formula Lemma 02TX. The identity holds by definition for $\alpha = [W]$ where W is an integral closed subscheme of X. For a general k-cycle α on X write $\alpha = \sum n_a [W_a]$ with $n_a \neq 0$, and $i_a : W_a \to X$ pairwise distinct integral closed subschemes. Then the family $\{W_a\}$ is locally finite on X. Set $P_a = \pi^{-1}(W_a) = \mathbf{P}(\mathcal{E}|_{W_a})$. Denote $i'_a : P_a \to P$ the corresponding closed immersions. Consider the fibre product diagram

$$\begin{array}{ccccc} P' & = & \coprod P_a & \xrightarrow{\coprod i'_a} & P \\ {\scriptstyle \pi'}\downarrow & & \downarrow{\scriptstyle \coprod \pi_a} & & \downarrow{\scriptstyle \pi} \\ X' & = & \coprod W_a & \xrightarrow{\coprod i_a} & X \end{array}$$

The morphism $p : X' \to X$ is proper. Moreover $\pi' : P' \to X'$ together with the invertible sheaf $\mathcal{O}_{P'}(1) = \coprod \mathcal{O}_{P_a}(1)$ which is also the pullback of $\mathcal{O}_P(1)$ is the projective bundle associated to $\mathcal{E}' = p^*\mathcal{E}$. By definition

$$c_j(\mathcal{E}) \cap [\alpha] = \sum i_{a,*}(c_j(\mathcal{E}|_{W_a}) \cap [W_a]).$$

Write $\beta_{a,j} = c_j(\mathcal{E}|_{W_a}) \cap [W_a]$ which is an element of $\mathrm{CH}_{k-j}(W_a)$. We have

$$\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_{P_a}(1))^i \cap \pi_a^*(\beta_{a,r-i}) = 0$$

for each a by definition. Thus clearly we have

$$\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_{P'}(1))^i \cap (\pi')^*(\beta_{r-i}) = 0$$

with $\beta_j = \sum n_a \beta_{a,j} \in \mathrm{CH}_{k-j}(X')$. Denote $p' : P' \to P$ the morphism $\coprod i'_a$. We have $\pi^* p_* \beta_j = p'_* (\pi')^* \beta_j$ by Lemma 02RG. By the projection formula of Lemma 02SU we conclude that

$$\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_P(1))^i \cap \pi^*(p_*\beta_j) = 0$$

Since $p_*\beta_j$ is a representative of $c_j(\mathcal{E}) \cap \alpha$ we win. □